

Northwind Robotics

Atlas AR-200 Autonomous Warehouse Robot — Operator Manual

Document revision 4.2, published March 2025. This manual covers the Atlas AR-200 model only. For the larger Orion OR-500 platform, refer to the separate Orion operator manual.

1. Overview

The Atlas AR-200 is a compact autonomous mobile robot designed for goods-to-person picking in warehouses up to 8,000 square metres. It was first released in 2023 and superseded the discontinued Atlas AR-100. The AR-200 navigates using a combination of LiDAR and floor-mounted QR fiducials.

Each Atlas AR-200 carries a maximum payload of 45 kilograms and travels at a top speed of 1.8 metres per second when carrying a load. Without a load it may travel at up to 2.4 metres per second. The robot has a turning radius of zero, as it uses independent wheel drive.

2. Technical Specifications

The AR-200 is powered by a 48-volt lithium-iron-phosphate battery pack with a capacity of 20 ampere-hours. A full charge takes approximately 90 minutes and provides up to 10 hours of continuous operation. The robot weighs 62 kilograms unladen. Its onboard compute runs on an ARM-based controller with 8 gigabytes of memory.

The operating temperature range is 5 to 40 degrees Celsius. The AR-200 is rated IP54 for dust and splash resistance and is not suitable for outdoor use or wash-down environments. Every unit carries a serial number of the form AR2-XXXXXX printed on a label beneath the battery tray.

3. Charging and Battery Care

The Atlas AR-200 returns to its charging dock automatically when its battery level falls below 15 percent. The docking contacts must be kept clean; wipe them weekly with a dry cloth. Do not use solvents on the charging contacts.

For long-term storage, charge the battery to approximately 60 percent and store the robot at a temperature between 10 and 25 degrees Celsius. Storing a fully charged or fully depleted battery for extended periods will shorten its service life. The battery pack is rated for 2,000 full charge cycles before its capacity drops below 80 percent.

To replace the battery, power down the robot using the recessed switch on the rear panel, wait 30 seconds, then release the battery tray by turning the two quarter-turn fasteners counter-clockwise. Replacement battery packs are sold under part number NW-BAT-48-20.

4. Safety

The AR-200 is fitted with front and rear safety scanners that stop the robot if a person or obstacle enters its protective field. The protective field extends 1.2 metres ahead of the robot at full speed. Never disable the safety scanners. The emergency stop button, coloured red, is located on the top rear surface of the robot and halts all motion immediately when pressed.

5. Error Codes

When the Atlas AR-200 encounters a fault, it displays an error code on its status ring and reports the same code to the fleet management console. The most common codes are listed below.

Code	Meaning	Operator action
E101	Battery temperature too high	Move robot to cooler area; wait 20 minutes
E203	Safety scanner obstructed	Clear the scanner window; wipe if dirty
E204	Safety scanner fault	Contact support; do not operate
E310	Lost floor fiducial	Return robot to a mapped aisle manually
E415	Payload exceeds limit	Remove excess load below 45 kg
E502	Drive motor overcurrent	Power-cycle; if persistent, contact support

If an error code not listed here appears, record the code and contact Northwind support at robotics-support@northwind-example.com. Quote your robot serial number when you make contact.

6. Routine Maintenance

Perform a visual inspection of the Atlas AR-200 at the start of each shift. Check that the safety scanner windows are clean, the wheels are free of debris, and the status ring illuminates green on power-up.

Every 500 operating hours, the drive wheels should be inspected for wear and replaced if the tread depth is below 1 millimetre. Drive wheels are sold in pairs under part number NW-WHL-200. The LiDAR sensor window should be cleaned monthly with a microfibre cloth and isopropyl alcohol.

A full preventive-maintenance service is recommended every 4,000 operating hours or once per year, whichever comes first. This service must be carried out by a Northwind-certified technician to keep the warranty valid.

7. Firmware

The Atlas AR-200 ships with firmware that can be updated over the air through the fleet management console. As of this revision, the current stable firmware version is 4.7.1. Firmware updates are applied automatically overnight unless the administrator has disabled automatic updates in the console settings. A firmware update takes about 12 minutes, during which the robot is unavailable.

8. Fleet Integration

Multiple Atlas AR-200 robots operate as a coordinated fleet under the Northwind fleet management console. The console assigns tasks, manages charging schedules, and resolves traffic conflicts between robots. A single console can manage up to 100 robots of mixed Atlas and Orion models.

Robots communicate with the console over a dedicated 5 gigahertz wireless network. Northwind recommends a separate wireless access point for every 1,000 square metres of floor space to maintain reliable coverage. If a robot loses its network connection, it will complete its current task and then stop safely in place, displaying error code E310 if it also loses its position reference.

9. Warranty Summary

The Atlas AR-200 is covered by Northwind's standard hardware warranty. Full warranty terms, including what is and is not covered, are set out in the Northwind Robotics Warranty and Support Policy, which takes precedence over this summary. In brief, the AR-200 carries a two-year warranty on the robot chassis and a one-year warranty on the battery pack.